CampusX

CLASSES SPECIFICATIONS DOCUMENT

Peer-to-Peer College Student Marketplace

Item	Description
Purpose	Defines properties, behaviour, and relationships for principal classes and components.
Architecture	Clean Architecture: Domain, Application, Infrastructure, and WebUI.
Specification basis	CampusX SRS and the existing Classes Specifications baseline. Confirm unresolved assumptions with the team.
Table format	Every class/component uses a Relations column alongside Details / responsibility; multiplicity is stated where supported.

Design note: multiplicities shown here reflect the current specification baseline, not a confirmed database schema. Optional user reporting, soft deletion, and final status values must be confirmed.

Domain Layer — Entities

ApplicationUser

Responsibility: Represents a student or administrator profile.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Stable identifier mapped to the Identity record.	Referenced by Product.SellerId, Report.ReporterId/TargetUserId, and ContactInquiry.BuyerId/SellerId.
Properties	FullName	string	Required display name.	Used by profile and admin DTOs; no direct entity relationship.
Properties	Email	string	Required email; validated by identity workflow.	Mapped to the authentication identity record.
Properties	PhoneNumber	string?	Optional phone number.	No direct entity relationship.
Properties	CollegeName	string	Required college name.	Used for profile display/filtering if supported.
Properties	ProfileImageUrl	string?	Optional profile image URL.	No direct entity relationship.
Properties	CreatedAt	DateTime	UTC creation time; private set.	No direct entity relationship.
Properties	IsBlocked	bool	Account block state; false initially.	Checked by authorization/account workflows.
Properties	Role	UserRole	Student or Admin.	Used by authorization policies; enum relationship.
Behaviour	UpdateProfile(...)	Method	Updates editable profile fields only.	Does not change Id, CreatedAt, or Role.
Behaviour	BlockAccount() / UnblockAccount()	Methods	Change account block state.	Used by admin workflows through IAdminService.

Product

Responsibility: A product listing offered by a student.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique product identifier.	Referenced by Report.TargetProductId and ContactInquiry.ProductId.
Properties	Title	string	Required product title.	Displayed in product summary/details DTOs.
Properties	Description	string	Required listing description.	No direct entity relationship.
Properties	Price	decimal	Must be greater than zero.	No direct entity relationship.
Properties	ImageUrl	string	Product image URL; requirement to confirm.	No direct entity relationship.
Properties	Status	ProductStatus	Available, Sold, Removed, or Archived (confirm final values).	Uses ProductStatus enum.
Properties	SellerId	Guid	Owner's user identifier.	Many Products belong to one ApplicationUser: ApplicationUser 1 : 0..* Product.
Properties	CategoryId	Guid	Product category identifier.	Many Products belong to one Category: Category 1 : 0..* Product.
Properties	CreatedAt / UpdatedAt	DateTime	Creation and last-update timestamps.	No direct entity relationship.
Properties	IsDeleted	bool	Soft-delete flag in current baseline; confirm.	Retains product references/history for reports and inquiries.
Behaviour	UpdateDetails(...)	Method	Updates editable listing details.	Seller authorization is checked by Application use case.
Behaviour	ChangeStatus(...)	Method	Changes listing state through allowed transitions.	Uses ProductStatus enum.

Category

Responsibility: Groups products for browsing and filtering.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique category identifier.	Referenced by Product.CategoryId.
Properties	Name	string	Required category name.	One Category has zero or many Products: Category 1 : 0..* Product.
Properties	Description	string?	Optional category description.	No direct entity relationship.
Properties	Products	Collection<Product>	Products in the category, if navigation property is used.	One-to-many with Product; can be represented by FK without navigation collection.
Behaviour	Rename(...)	Method	Changes category name after validation.	Category management use case only if included in final requirements.

Report

Responsibility: A report submitted about a product or optionally a user.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique report identifier.	Referenced by admin review workflows.
Properties	ReporterId	Guid	User who submitted the report.	Many Reports belong to one ApplicationUser: ApplicationUser 1 : 0..* Report.
Properties	TargetProductId	Guid?	Product being reported; required by baseline.	Many Reports may target one Product: Product 1 : 0..* Report.
Properties	TargetUserId	Guid?	Optional user target; confirm requirement.	If enabled, many Reports may target one ApplicationUser: ApplicationUser 1 : 0..* Report.
Properties	Reason	string	Required reason for report.	No direct entity relationship.
Properties	Status	ReportStatus	Pending, Reviewed, or other agreed values.	Uses ReportStatus enum.
Properties	CreatedAt	DateTime	UTC submission timestamp.	No direct entity relationship.
Behaviour	Review(...)	Method	Records moderation decision/status.	Admin workflow; exactly one target should be set if user reporting is supported.

ContactInquiry

Responsibility: A buyer-to-seller inquiry about a product; not a multi-message chat.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique inquiry identifier.	Referenced by inquiry service and response DTOs.
Properties	ProductId	Guid	Product being queried.	Many inquiries may reference one Product: Product 1 : 0..* ContactInquiry.
Properties	BuyerId	Guid	Student who sends inquiry.	Many inquiries may belong to one buyer: ApplicationUser 1 : 0..* ContactInquiry.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	SellerId	Guid	Product owner / recipient.	Many inquiries may belong to one seller: ApplicationUser 1 : 0..* ContactInquiry.
Properties	Message	string	Initial inquiry message.	No direct entity relationship.
Properties	CreatedAt	DateTime	UTC creation timestamp.	No direct entity relationship.
Behaviour	ValidateParticipants()	Method	Ensures buyer/seller are valid and seller owns the product.	Coordinates Product and ApplicationUser references.

Domain Layer — Enums

UserRole

Responsibility: Supported user roles.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Student	enum value	Standard marketplace user.	Used by ApplicationUser.Role and authorization.
Properties	Admin	enum value	Administrator role.	Used by ApplicationUser.Role and admin authorization.

ProductStatus

Responsibility: Lifecycle state for product listings.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Available	enum value	Listing is available.	Used by Product.Status and search filters.
Properties	Sold	enum value	Listing has been sold.	Used by Product.Status.
Properties	Removed	enum value	Listing is removed from public view.	Used by Product.Status.
Properties	Archived	enum value	Optional archived state; confirm.	Used by Product.Status if approved.

ReportStatus

Responsibility: Lifecycle state for reports.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Pending	enum value	Awaiting moderation.	Used by Report.Status.
Properties	Reviewed	enum value	Reviewed by admin.	Used by Report.Status; additional outcomes need agreement.

Application Layer — Service Contracts

IAuthService

Responsibility: Authentication use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	RegisterAsync(RegisterStudentDto)	Operation	Registers a student and returns a safe result.	Uses RegisterStudentDto and AuthResponseDto; delegates identity operations to IIdentityService.
Behaviour	LoginAsync(LoginDto)	Operation	Authenticates credentials without exposing secrets.	Uses LoginDto and AuthResponseDto; depends on IIdentityService.
Behaviour	LogoutAsync()	Operation	Ends the current authentication session.	Implemented through identity/authentication adapter.

IProfileService

Responsibility: Profile read and update use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	GetProfileAsync(userId)	Operation	Returns the current user's profile.	Reads ApplicationUser via IUserRepository and maps to UserProfileDto.
Behaviour	UpdateProfileAsync(userId, dto)	Operation	Validates and updates editable profile fields.	Uses UpdateProfileDto, ApplicationUser, IUserRepository, and UserProfileDto.

IProductService

Responsibility: Product listing and search use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	CreateAsync(sellerId, dto)	Operation	Validates and creates a listing.	Uses CreateProductDto, Product, IProductRepository, ICategoryRepository.
Behaviour	UpdateAsync(productId, actorId, dto)	Operation	Updates listing only for an authorized seller/admin.	Uses UpdateProductDto, Product, IProductRepository.
Behaviour	GetDetailsAsync(productId)	Operation	Returns product details.	Maps Product and related seller/category data to ProductDetailsDto.
Behaviour	SearchAsync(filter)	Operation	Searches and pages listings.	Uses ProductSearchFilterDto, IProductRepository, PagedResult<ProductSummaryDto>.

IInquiryService

Responsibility: Buyer-to-seller inquiry use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	CreateAsync(buyerId, dto)	Operation	Validates participants and product ownership.	Uses CreateInquiryDto, Product, ApplicationUser, IContactInquiryRepository.
Behaviour	GetForUserAsync(userId)	Operation	Returns inquiries visible to a participant.	Maps ContactInquiry to InquiryResponseDto; restricts to buyer/seller.

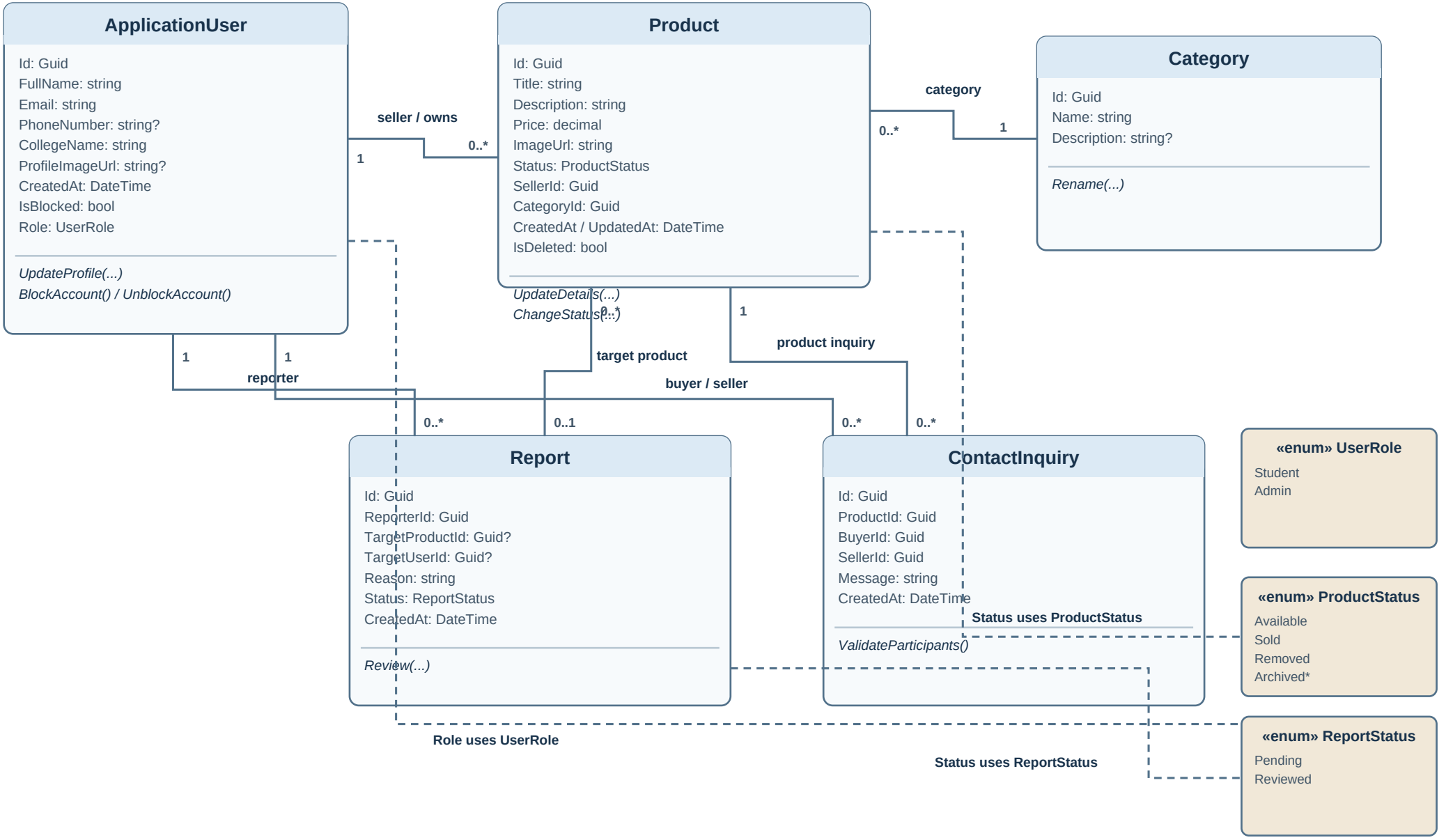
IAdminService

Responsibility: Administrative moderation and management use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	SearchUsersAsync(filter)	Operation	Searches users for admin review.	Uses UserSearchFilterDto, IUserRepository, AdminUserDto.
Behaviour	ManageProductAsync(...)	Operation	Performs approved admin product actions.	Uses ProductAdminFilterDto/AdminProductDto and IProductRepository.
Behaviour	ReviewReportAsync(reportId, decision)	Operation	Reviews a report and records outcome.	Uses Report, IReportRepository, ReportStatus.

CampusX — UML Class Diagram

Core Domain model • Properties, behaviours, relationships, and multiplicities



* Archived is optional in the current baseline. Nullable report targets and final status values must be confirmed against the SRS.

Clean Architecture dependency direction: WebUI → Application → Domain. Infrastructure implements Application abstractions and depends inward on Application/Domain. Domain has no dependency on outer layers.